

1.6: Absolute Value Functions

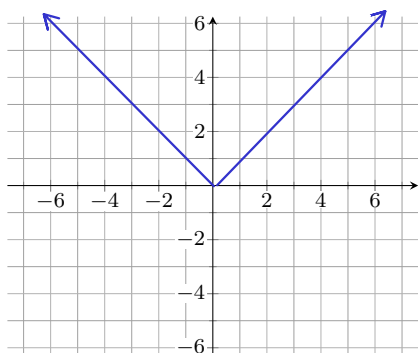
Definition. (Absolute Value Function)

The absolute value function can be defined as a piecewise function

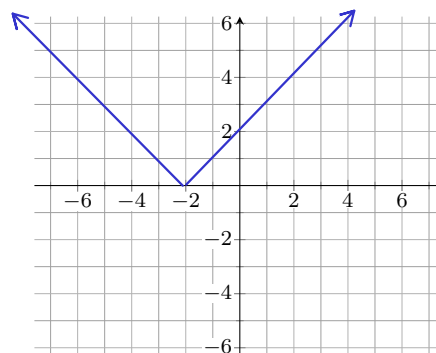
$$f(x) = |x| = \begin{cases} x, & \text{if } x \geq 0 \\ -x, & \text{if } x < 0 \end{cases}$$

Example. Graph the following

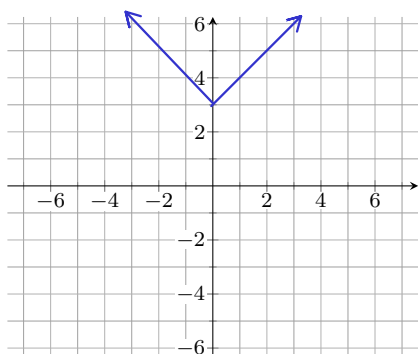
$$f(x) = |x|$$



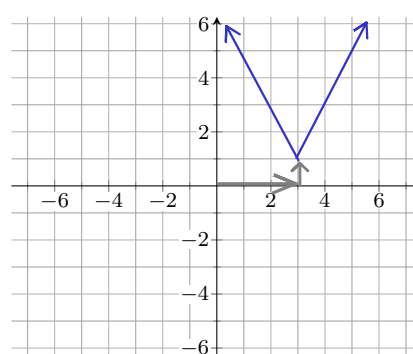
$$f(x) = |x - 2|$$



$$f(x) = |x| + 3$$



$$f(x) = 2|x - 3| + 1$$



Example. Solve the following equations

Graphs

$$|x| = 3$$

$$x = \pm 3$$

$$|2x + 6| = 2$$

$$2x + 6 = \pm 2$$



$$2x + 6 = -2$$

$$2x = -8$$

$$x = -4$$

$$2x + 6 = 2$$

$$2x = -4$$

$$x = -2$$

$$x = \frac{-6 \pm 2}{2}$$

$$|x + 3| - 1 = 0$$

$$|x + 3| = 1$$

$$x + 3 = \pm 1$$

$$x = -3 \pm 1$$



$$x = -4$$

$$x = -2$$

$$2 - |4 - x| = 0$$

$$2 = |4 - x|$$

$$\pm 2 = 4 - x$$

Redundant

$$x = 4 - (\pm 2)$$



$$x = 2$$

$$x = 6$$

Example. Solve the following inequalities. Give your answer in interval notation. [Graphs](#)

$$|x - 5| \leq 4$$

$$-4 \leq x - 5 \leq 4$$

$$1 \leq x \leq 9$$

$$[1, 9]$$

$$\left| \frac{x}{4} - 1 \right| < 2$$

$$-2 < \frac{x}{4} - 1 < 2$$

$$-1 < \frac{x}{4} < 3$$

$$-4 < x < 12$$

$$(-4, 12)$$

$$|2 - x| \leq 3$$

$$-3 \leq 2 - x \leq 3$$

$$-5 \leq -x \leq 1$$

$$-1 \leq x \leq 5$$

$$[-1, 5]$$

$$|x - 3| > 2$$

$$\swarrow \quad \searrow$$

$$x - 3 > 2$$

$$x > 5$$

$$-(x - 3) > 2$$

$$x - 3 < -2$$

$$x < -5$$

$$\swarrow \quad \searrow$$

$$(-\infty, -5) \cup (5, \infty)$$